Lab Report: Principal Component Analysis (PCA)

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The objective of PCA is to condense the information present in the columns of a dataset into a reduced number of principal components, thereby facilitating the interpretation of differences between individuals from complex data. In this lab, we propose to apply this method to the `crabs` and `DonneesSNPnormalisées.RData` datasets and interpret the results.

1. Course Reminders and Theoretical Foundations

1.1. The case of an individual's grades

Let us explain the principle of PCA using the example seen in the course. Consider a row vector $X \in \mathbb{R}^{1\times p}$ of p random variables, whose coordinates represent a student's averages in p distinct subjects.

$$X = [X_1, \dots, X_p] \in \mathbb{R}^p \tag{1}$$

We assume that the X_i are centred random variables, i.e. $\mathbb{E}[X_1] = \cdots = \mathbb{E}[X_p] = 0$. Our goal is to represent X with a new vector $Z \in \mathbb{R}^{1\times p}$ of p new random variables $Z_1, \dots, Z_p$. In other words, there exists a change-of-basis matrix $P \in \mathbb{R}^{p\times p}$, whose i-th column contains the coordinates of Z_i in the original basis. P is then a change-of-basis matrix such that:

$$Z = [Z_1, \dots, Z_p] = [X_1, \dots, X_p]P \tag{2}$$

The variables $Z_1, \dots, Z_p$ must satisfy the following two constraints:

- $\forall i \in [\![1, p]\!]$, Z_i is a linear combination of $X_1, \dots, X_p$.
- The Z_i are pairwise uncorrelated, i.e. for all $i \neq j$,

$$\begin{aligned} \mathbb{C}\mathrm{ov}(Z_i, Z_j) &= \mathbb{E}[Z_i Z_j] - 0 \quad \text{(by König-Huygens and centred r.v.)} \\ &= 0. \end{aligned} \tag{3}$$

In matrix form, the second condition is written by cancelling all off-diagonal covariance products, retaining only the variance terms:

$$\begin{aligned} \mathbb{C}\mathrm{ov}(Z) &= \mathbb{E}[Z^T Z] - 0 \quad \text{(by König-Huygens and centred r.v.)} \\ &= \begin{pmatrix} \mathbb{E}[Z_1^2] & 0 & \cdots & 0 \\ 0 & \mathbb{E}[Z_2^2] & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbb{E}[Z_p^2] \end{pmatrix} \end{aligned} \tag{4}$$

1.2. Generalisation to n individuals

We now wish to generalise this principle by considering the grade records of n students. The components X_i of X become vectors $X_{\cdot i} = [X_{1i}, \dots, X_{ni}]^T \in \mathbb{R}^{n\times 1}$, the grades of the class in the i-th subject. Thus, $X = [X_1, \dots, X_p] \in \mathbb{R}^p$ becomes $X = [X_{\cdot 1}, \dots, X_{\cdot p}] \in \mathbb{R}^{n\times p}$. Concretely, X now represents the p grades of n individuals.

$$X = \begin{pmatrix} X_{11} & \dots & X_{1p} \\ \vdots & \ddots & \vdots \\ X_{n1} & \dots & X_{np} \end{pmatrix} \tag{5}$$

where X_{ij} is the grade of the i-th student in the j-th subject. Our objective remains the same: we wish to represent the grade record in another basis, i.e. with a vector $Z \in \mathbb{R}^{n \times p}$. In other words, we would like there to exist a change-of-basis matrix $P \in \mathbb{R}^{p \times p}$ such that:

$$Z = XP \tag{6}$$

which satisfies the following generalised constraints:

- $\forall i \in [\![1, p]\!]$, $Z_{\cdot i}$ is a linear combination of $X_{\cdot 1}, \dots, X_{\cdot p}$.
- The $Z_{\cdot i}$ are pairwise uncorrelated:

$$\begin{aligned} \mathbb{C}\mathrm{ov}(Z) = \mathbb{E}(Z^T Z) &= \mathbb{E}[(XP)^T XP] \\ &= \mathbb{E}[P^T X^T X P] \\ &= P^T \mathbb{E}[X^T X] P \text{ is a diagonal matrix} \end{aligned} \tag{7}$$

We then make an empirical approximation of the expectation, since the distribution of the random variables is unknown. The second condition then becomes:

$$\exists (\lambda_1, \dots, \lambda_p) \in \mathbb{R}^p \quad \text{such that} \quad Z^T Z = P^T \left(\frac{1}{n} X^T X \right) P = \mathrm{Diag}(\lambda_1, \dots, \lambda_p) \tag{8}$$

It is legitimate to ask whether such a matrix P exists. P does indeed exist: $X^T X$ is a real symmetric matrix and is therefore diagonalisable by an orthogonal change-of-basis matrix, in accordance with the spectral theorem.

1.3. How to interpret a PCA?

In the context of PCA, the selection of vectors for the new basis rests on a fundamental principle: maximising the explained variance. This criterion is based on two observations. A vector with a small variance tends to be deterministic, contributing little new information to the analysis; conversely, a vector with a high variance is likely to contain significant information about the variability of the data.

The procedure for selecting principal components therefore aims to optimise a distance criterion between the different vector subspaces of the diagonalised matrix. This approach allows the most informative components to be retained, i.e. those that best capture the variability inherent in the original data.

To interpret the new variables, we analyse the contributions of the old variables in constructing the new ones. The correlation circle is a visual tool that intuitively represents the correlation between the old and new bases. The principle consists in projecting the old variables onto the new basis to facilitate this interpretation. For all $i, j \in [\![1, p]\!]$:

$$\exists (\alpha_1, \dots, \alpha_p) \in \mathbb{R}^p \quad \text{such that} \quad X_j = \alpha_1 Z_1 + \dots + \alpha_p Z_p$$

Thus, the projection of the X_i onto the Z_i is the covariance between the X_i and the Z_i:

$$\begin{aligned} \mathbb{C}\mathrm{ov}(Z_i, X_j) &= \mathbb{C}\mathrm{ov}\left(Z_i, \sum_{k=1}^{p} \alpha_k Z_k \right) \\ &= \sum_{k=1}^{p} \alpha_k \mathbb{C}\mathrm{ov}(Z_i, Z_k) \\ &= \alpha_j \lambda_j \quad (\text{since } \mathbb{C}\mathrm{ov}(Z_i, Z_k) = 0 \text{ if } i \neq k \text{ and } \mathbb{C}\mathrm{ov}(Z_j, Z_j) = \lambda_j) \end{aligned}$$

2. Crabs

This study concerns a dataset called `crabs`, containing information on 200 crab specimens. Each specimen is characterised by eight variables, five of which are quantitative. Our PCAs will focus exclusively on these quantitative variables.

Variable	Description
FL	Frontal lobe width
RW	Rear width of the carapace
CL	Carapace length
CW	Carapace width
BD	Body depth of the crab

Table 1: Quantitative variables of the `crabs` dataset

2.1. Naive approach: PCA without adjustments

We applied PCA directly to the dataset without any specific pre-processing. To do this, we used the `PCA` function from the Scikit-learn library, standardising the data using the `scale` method.

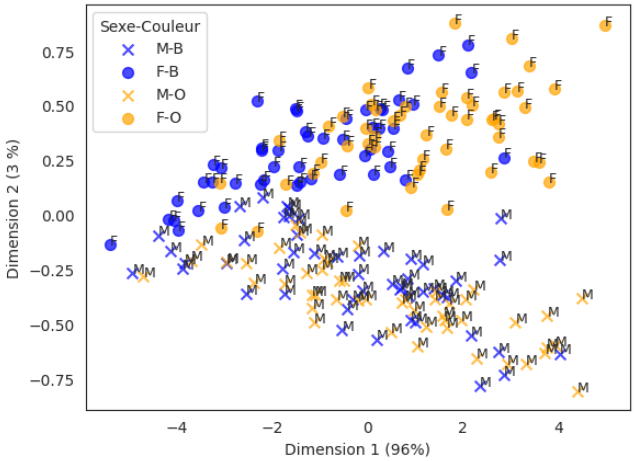

Figure 1: Distribution of crabs by sex and colour in the plane of the first two principal components of the PCA without adjustments.

The second principal component allows a relatively clear distinction between males and females. We will see later that this component appears to be mainly linked to RW, i.e. the rear width of the carapace. However, the first principal component does not seem to effectively discriminate crabs by colour or sex. Let us then analyse the distribution of explained variance in order to interpret the new variables.

	Dimension	Variance expliquée	% variance expliquée	% cum. var. expliquée
0	Dim1	4.812899	96.0	96.0
1	Dim2	0.152447	3.0	99.0
2	Dim3	0.046867	1.0	100.0
3	Dim4	0.011191	0.0	100.0
4	Dim5	0.001720	0.0	100.0

Figure 2: Distribution of explained variance of the PCA without adjustments.

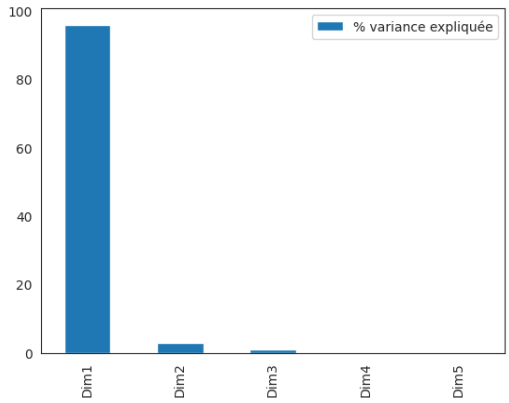

Figure 3: Bar chart of the explained variance distribution of the PCA without adjustments: 96% for the first component, 3% for the second, 1% for the third, the others being negligible.

The first principal component alone explains 96% of the total variance. This result suggests a strong correlation between most of the measured variables, potentially linked to a common latent variable which is the size of the crab: there is therefore a **size effect**! Let us display the correlation circle to clarify our hypothesis.

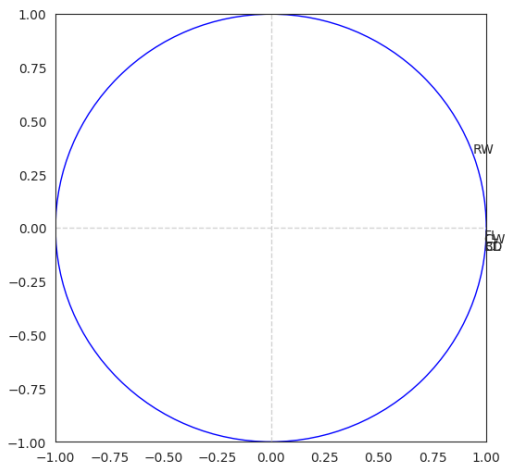

Figure 4: Correlation circle of the PCA without adjustments in the plane of the first two components.

With the exception of the variable RW, all other variables show a strong correlation with the first principal component, suggesting that they vary similarly and are probably linked to the overall size of the crab, since they measure different parts of its body. Thus, the first component only serves to differentiate small from large crabs. However, the variable RW, corresponding to the rear width of the carapace, stands out with a slightly different orientation, indicating a variation potentially independent of the other biological characteristics.

The results highlight significant sexual dimorphism, especially in pelvis width, which appears greater in females. The strong correlation between the majority of variables and the first principal component suggests the existence of a dominant latent variable linked to the overall size of the crab. The relative independence of the RW variable could reflect a specific morphological characteristic influenced by factors other than general growth.

2.2. First method to correct the size effect: using components 2 and 3 of the PCA.

To eliminate the size effect, a first straightforward method is to use the results of the PCA performed previously. We then project the individuals onto the plane formed by components 2 and 3 of this PCA. This yields two variables with well-distributed variances, indicating the absence of a latent variable.

However, these variances remain low, meaning they exhibit little variability, hence a certain constancy among all individuals, and therefore that these are not discriminating components.

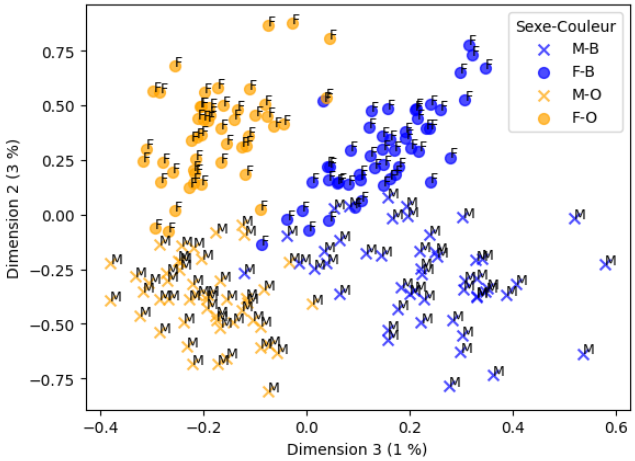

Figure 5: Distribution of crabs by sex and colour in the plane of principal components 2 and 3 of the PCA without adjustments.

Component 2, linked to the rear width of the carapace, still allows males and females to be well distinguished in our analysis. Furthermore, component 3 appears to separate crabs according to their colour, thus distinguishing blue crabs from orange crabs. This component contributes in particular to differences in carapace size, even if the distinction between the two groups is not perfectly clear. It is possible that other techniques more sophisticated than PCA could better classify these groups, or that this variability reflects the complexity of nature, where strict generalisations are difficult to make. For example, there could be blue crabs with small carapaces, which nuances this separation. Finally, let us also examine the correlation circle to better understand the relationships between morphological variables.

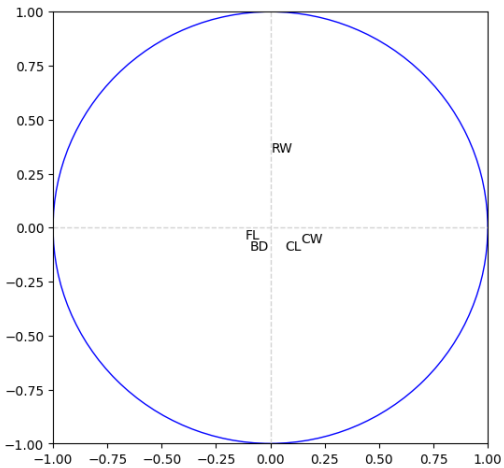

Figure 6: Correlation circle of the PCA without adjustments in the plane of components 2 and 3.

Component 3 retains characteristics related to size, but these are not correlated with the overall size of the crabs given the 1% explained variance. This means that even if component 3 contains information on certain morphological dimensions, it is not dominated by overall size. However, the vectors of component 3 are close to the centre, showing that they have a low explained variance. This component reveals that blue crabs have a carapace that is both wider and longer, while their frontal lobe and body depth are smaller. Conversely, for orange crabs, the carapace is smaller, but their frontal lobe and body depth are greater. This interpretation is supported by the direction of the arrows in the biplot: the variables BD and FL point to the left, indicating that they contribute negatively to component 3, while CL and CW point to the left, showing that they contribute positively to it.

This component seems to reflect a ratio between carapace size (CL, CW) and body depth/frontal lobe width (BD, FL). We could interpret it as a measure of the crab's "shape", independently of its overall size.

2.3. Second method to correct the size effect: normalising the data by the most correlated variable.

Another technique to eliminate the size effect consists in identifying the variable most strongly correlated with the others in the correlation matrix, which we assume to be the latent variable linked to overall size. Once this variable is identified, we divide all other variables by it in order to eliminate the correlation with size. By doing so, we no longer consider absolute values but size ratios, which allows us to better isolate relative morphological differences between individuals. Let us display the variable correlation matrix.

	FL	RW	CL	CW	BD
FL	1.000000	0.906988	0.978842	0.964956	0.987627
RW	0.906988	1.000000	0.892743	0.900402	0.889205
CL	0.978842	0.892743	1.000000	0.995023	0.983204
CW	0.964956	0.900402	0.995023	1.000000	0.967812
BD	0.987627	0.889205	0.983204	0.967812	1.000000

Figure 7: Correlation matrix of the quantitative variables of `crabs`

The column with the strongest correlation with the others is carapace length (CL). Consequently, as indicated above, we divide all columns by CL, then remove the latter. Finally, we project the individuals onto the plane defined by the first two components of the normalised PCA.

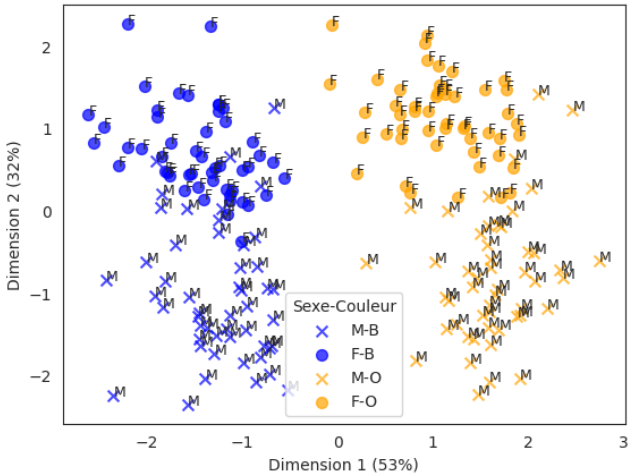

Figure 8: Distribution of crabs by sex and colour in the plane of the first two principal components of the normalised PCA.

Component 1 allows blue and orange crabs to be distinguished. As we will see on the correlation circle, it is strongly correlated with body depth. Crabs with a large body depth are mainly orange, while those with a smaller depth are blue. Component 2, for its part, allows males and females to be differentiated. On the correlation circle, we will observe that it is correlated with the rear width of the crabs, with females having a greater rear width than males. Finally, let us visualise the explained variances.

	Dimension	Variance expliquée	% variance expliquée	% cum. var. expliquée
0	Dim1	2.125923	53.0	53.0
1	Dim2	1.304129	32.0	85.0
2	Dim3	0.327023	8.0	93.0
3	Dim4	0.263025	7.0	100.0

Figure 9: Distribution of explained variance of the PCA without adjustments.

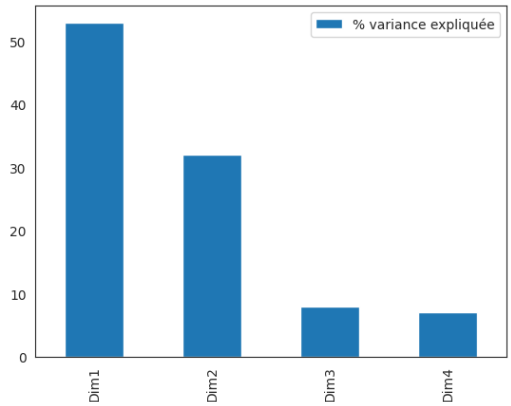

Figure 10: Bar chart of the explained variance distribution of the normalised PCA: 53% for the first component, 32% for the second, 8% for the third, and 4% for the fourth.

There is no size effect; the components are balanced in explained variance and present sufficient variability to contain information. We will display the correlation circle to visualise these relationships.

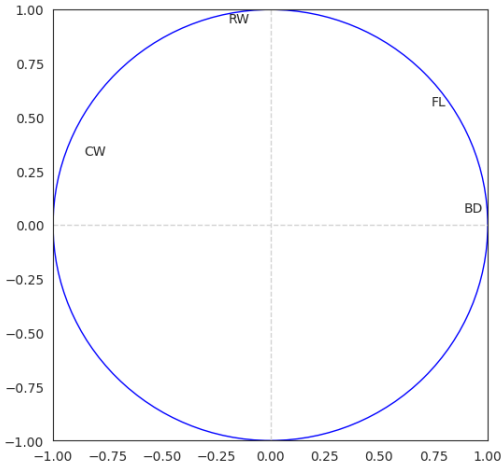

Figure 11: Correlation circle of the normalised PCA in the plane of components 1 and 2.

Component 1 is primarily associated with the relative body depth $\frac{BD}{CL}$, which can be interpreted as an indicator of the crabs' "stockiness". In parallel, component 2 is linked to the relative pelvis width $\frac{RW}{CL}$. CW points towards the upper left corner, suggesting a tendency towards large RW and small BD, characteristics of blue females. Conversely, a large FL, combined with high RW and BD, is typical of orange females. Thus, crabs with larger proportions are generally females.

3. Single Nucleotide Polymorphism

We used a genetic dataset from the file `DonneesSNPnormalisées.RData`, which we converted to CSV format in order to use it in Python. This dataset contains 728 rows representing individuals and 1102 features. The columns can be classified as follows:

- **1 identifier column**: Each individual is uniquely identified by an identifier column.
- **5500 raw SNP columns**: These columns correspond to data on SNPs (Single Nucleotide Polymorphisms). The values in these columns are between 0, 1 and 2, representing the number of specific alleles relative to a reference:
  - **0**: Absence of the reference allele at a given position.
  - **1**: Presence of one reference allele and one other allele.
  - **2**: Presence of two reference alleles at this position.

  However, these columns contain missing values (NaN) and have not been pre-processed.
- **5500 normalised SNP columns**: In addition to the raw columns, we have 5500 corresponding columns containing the pre-processed (normalised) SNPs, which we use for our analyses.
- **1 `origin` column**: This column contains an additional feature indicating the origin of each individual.

The analysis was focused on the **normalised SNPs**, as they are already pre-processed and ready for statistical analysis. Since the number of features exceeds that of individuals, we use Python PCA functions adapted to this particularity.

3.1. Naive approach: PCA without adjustments

We applied PCA directly to the normalised dataset without any specific pre-processing. To do this, we used the `PCA` function from the Scikit-learn library.

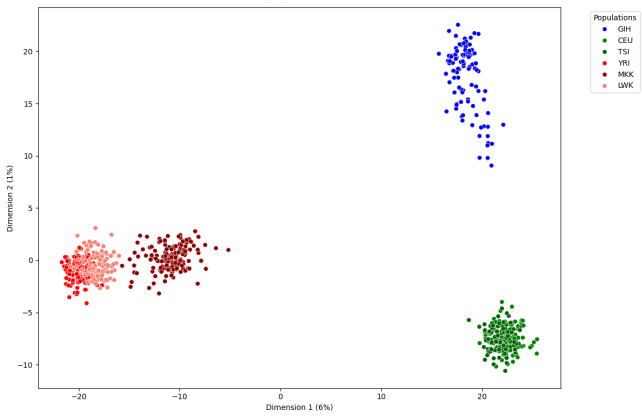

Figure 12: Distribution of SNPs by ethnic origin in the plane of the first two principal components of the PCA without adjustments.

The separation of populations is clear in the plane of the first two principal components, especially given the low explained variance. This suggests that even with a small fraction of the total variance, we manage to obtain very distinct populations, or that despite the existence of different populations, humans are genetically almost identical.

We will see in the next section that each population has specific reference SNPs. For example, the MKK tend towards 0, suggesting they probably have only one African reference allele. In contrast, the YRI and LWK appear to have two African reference SNPs, as their component on dimension 1 is strongly negative. Let us analyse the explained variances in more detail.

	Dimension	Variance expliquée	% variance expliquée	% cum. var. expliquée
0	Dim1	3.472927e+02	6.33	6.33
1	Dim2	6.186228e+01	1.13	7.46
2	Dim3	4.657875e+01	0.85	8.30
3	Dim4	1.955498e+01	0.36	8.66
4	Dim5	1.827538e+01	0.33	8.99
...	...	...	...	...
723	Dim724	1.722579e+00	0.03	99.94
724	Dim725	1.649286e+00	0.03	99.97
725	Dim726	1.630234e+00	0.03	100.00
726	Dim727	1.061389e-01	0.00	100.00
727	Dim728	9.131385e-30	0.00	100.00

Figure 13: Distribution of explained variance of the PCA without adjustments.

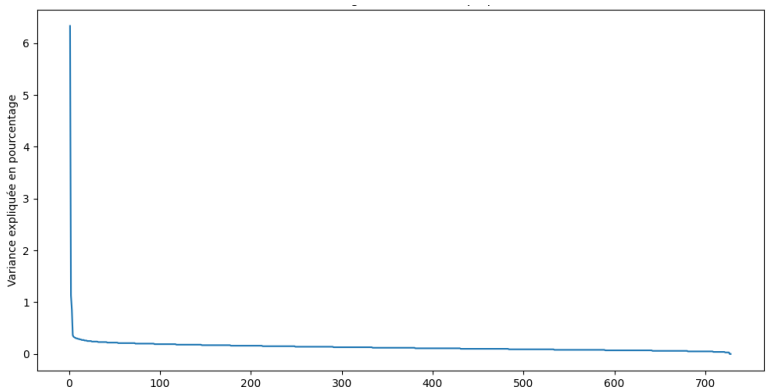

Figure 14: Explained variance of the PCA without adjustments as a function of component: 6.33% for the first component, 1.13% for the second, 0.85% for the third, and the rest is negligible.

On the one hand, we note that the PCA returns only 728 components. Indeed, even though we have 5500 columns, the maximum number of principal components is limited by the number of individuals, i.e. 728 in this case. This is explained by the fact that PCA is based on the decomposition of the data covariance matrix. This matrix has a dimension equal to the minimum between the number of individuals and the number of variables.

On the other hand, the variance explained by the first two components (6% and 1.13%) is low relative to the total number of features, but those of the other components are even lower. This is not unusual in genetic analyses with a large number of SNPs. The absence of a dominant size effect suggests that genetic variation is distributed across many SNPs, almost all identical to one another, and that only a small group of varying genes explains the differences.

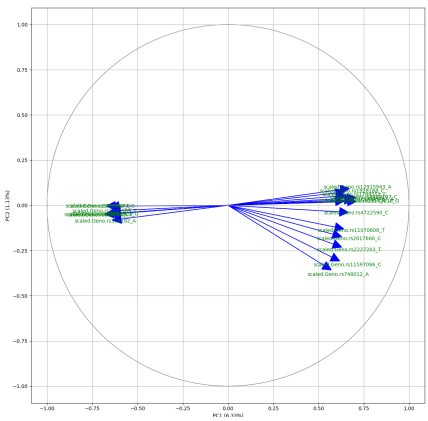

Figure 15: Correlation circle of the PCA without adjustments in the plane of the first two components. Only the 20 vectors with the largest norm have been shown for readability.

A SNP pointing to the right on component 1 could indicate that Eurasian populations more often have values of 1 or 2, while African populations tend towards 0, and vice versa for a SNP pointing to the left. Thus, reference SNPs with positive values on PC1 are probably more frequent in Eurasia, and those with negative values in Africa. SNPs deviating downwards on component 2 could distinguish European populations from Indian populations. These SNPs are more present in Europeans than in Indians.

Component 1 (6% of variance) could represent a "gradient of genetic diversity" between Africa and Eurasia, reflecting the history of human migration out of Africa. Component 2 (1% of variance) appears to capture differences between Europeans and Indians, probably linked to local adaptations or genetic drift effects following this migration. At this stage, we could hypothesise that the MKK represent North African populations.

3.2. PCA on the plane spanned by components 2 and 3.

Let us now project the individuals from our dataset onto the plane formed by components 2 and 3 of the same PCA.

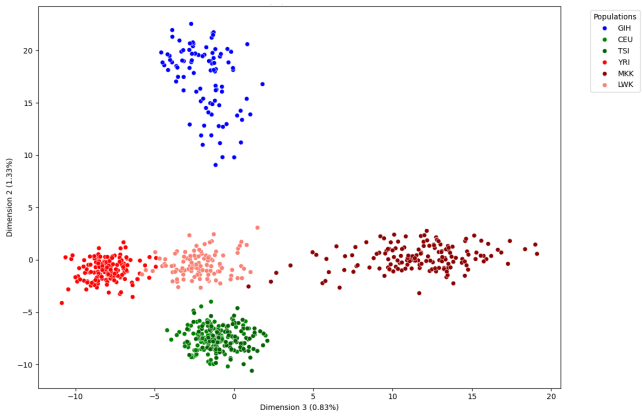

Figure 16: Distribution of SNPs by ethnic origin in the plane of components 2 and 3 of the PCA without adjustments.

The separation of populations remains very marked on components 2 and 3. Component 3 appears to better differentiate the various African populations. The vertical axis continues to distinguish Indians, Africans and Europeans. We will now examine the correlation circle to identify the SNPs that help differentiate populations.

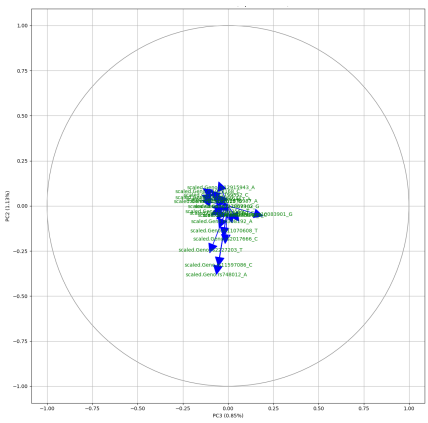

Figure 17: Correlation circle of the PCA without adjustments in the plane of components 2 and 3. Only the 20 vectors with the largest norm have been shown for readability.

We can make analogous interpretations to section 3.1 regarding the separation of the different African

populations: component 3 appears to distinguish the various African populations. Nevertheless, we observe a major difference: most SNPs are grouped near the centre, indicating a low individual contribution to these components, due to the very low explained variance of components 2 and 3.

4. Conclusion

This lab allowed us to explore Principal Component Analysis through two datasets: `crabs` and `Single Nucleotide Polymorphisms`. These analyses highlighted the power of PCA to reduce data dimensionality while preserving the most relevant information:

- In `crabs`, we were able to observe how PCA can reveal hidden structures in the data, notably by highlighting morphological differences linked to the sex and colour of the crabs. We also explored different methods to correct the size effect, which allowed for a better distinction of morphological characteristics independent of the individuals' overall size. For the SNP analysis, PCA demonstrated its ability to handle very high-dimensional data, enabling us to visualise and interpret genetic differences between populations that would have been difficult to perceive otherwise.
- It is important to note that in this lab, we limited ourselves to 2D representations for simplicity and readability. However, PCA also allows 3D representations and even higher dimensions, which can sometimes reveal even more complex structures in the data.

PCA finds applications in both unsupervised and supervised learning:

- In unsupervised learning, as we have seen in this lab, PCA allows latent structures to be discovered in data without prior information. It can be used for visualisation, noise reduction, or data compression.
- In supervised learning, PCA can be used as a pre-processing step to reduce data dimensionality before applying a classification or regression algorithm.

In conclusion, PCA proves to be a powerful and versatile tool in data analysis, capable of revealing hidden structures in complex, high-dimensional datasets. Its utility extends well beyond simple visualisation, playing a crucial role in many machine learning and exploratory data analysis applications.